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The Development Lengths of Laminar Pipe and Channel Flows

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The authors' research work into fully developed pulsating and oscillating laminar pipe and channel flows raised questions regarding the development length of the corresponding steady flow. For this development length, i.e., the distance from the entrance of the pipe to the axial position where the flow reaches the parabolic velocity profile of the Hagen-Poiseuille flow, a wide range of contradictory data exists. This is shown through a short review of the existing literature. Superimposed diffusion and convection, together with order of magnitude considerations, suggest that the normalized development length can be expressed as $L/D=C_0+C_1 Re$ and for $Re\to 0$ one obtains $C_0=0.619$, whereas for $Re\to\infty$ one obtains $C_1=0.0567$. This relationship is given only once in the literature and it is presumed to be valid for all Reynolds numbers. Numerical studies show that it is only valid for $Re\to 0$ and $Re\to\infty$. The development length of laminar, plane channel flow was also investigated. The authors obtained similar results to those for the pipe flow: $L/D=C_0'+C_1'$; Re, where $C_0'=0.631$ and $C_1'=0.044$. Finally, correlations are given to express L/D analytically for the entire Re range for both laminar pipe and channel flows.
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1 Introduction and Aim of Work

Two of the classical flows treated in introductory fluid mechanics lectures are laminar, fully developed pipe flow, in other words, the so-called Hagen-Poiseuille flow, and the corresponding plane channel flow. The development of these flows from preassigned velocity profiles require certain axial distances from the pipe or the channel inlet. Irrespective of a particular inlet velocity profile or what happens in detail at the entrance of a pipe or a channel, the physical mechanisms behind such axial development of a flow are well established and understood. Owing to the no-slip velocity condition at walls, the fluid next to the wall is immediately slowed as soon as the flow enters a pipe or a channel. This retardation near the wall spreads inwards owing to viscous effects and the slowed-down fluid close to the wall causes the fluid in the center to move faster, since the cross-sectional mass flow rate at any axial location remains constant. Ultimately, moving in the flow direction, the fully developed state of the flow is reached, i.e., the parabolic velocity distribution of the Hagen-Poiseuille flow develops. The closest location from the entrance where this phenomenon occurs defines the hydrodynamic entrance length as the distance from the inlet of the pipe to the location of the fully developed pipe flow or the corresponding position of the fully developed channel flow.

As the above explanations show, the phenomenon of laminar pipe and channel flow development is fully understood and the dependence of the development length on the Reynolds number, when derived by the type of considerations given above, must have two terms, due to the superposition of the convection and diffusion velocities. However, in spite of this, there are derivations in the literature that yield $L/D=C\,\mathrm{Re}$. This is readily illustrated in Sec. 2, where a brief literature survey is provided, indicating that there have been numerous investigations of the phenomena. Unfortunately no clear and final analysis or experimental and numerical studies exist that state precisely what the values of the constants C_0 and C_1 in the L/D-relationship should be and also whether the simple scale analysis given above is valid for all Reynolds numbers. Hence it may be concluded that the development of laminar pipe or channel flow is not correctly described in the literature. The present study was carried out in order to remedy this situation and also to provide quantitative information on the ratio of the development length to pipe diameter or channel height as a function of Reynolds number.

Owing to the situation outlined in Sec. 2, it was felt that a detailed study is needed to finalize the investigations on developing laminar pipe flow and the corresponding channel flow that eventually leads to the fully developed velocity distribution. Detailed studies in this respect were carried out, and the final results are presented in this paper. Although these investigations were performed numerically, experimentally and analytically, only the numerical part of the study was found to be really successful. The numerical studies resulted in reliable data that were evaluated and the final results are presented in Sec. 3. The experimental studies are summarized in Sec. 4 together with problems encountered also during analytical investigations of the development length of laminar pipe and channel flows. Conclusions and final remarks are provided in Sec. 5.

2 Summary of Results of Previous Investigations

There have been numerous investigations of the development length for laminar pipe flows, carried out not only using various approximate analytical methods, but also by performed numerical and experimental investigations. Good summaries of the work in the past are provided in Schlichting [1], giving special references to the publications of Schiller [2], Langhaar [3], and Sparrow [4]. Additional publications are referenced by Schmidt and Zeldin [5], followed in further studies by Lew and Fung [6]. A few years later, the development length of laminar pipe flows was looked at by Mohanty and Asthana [7]. All these publications indicate that the phenomenon of laminar flow development in the entrance region of pipe flows has attracted considerable attention in fluid mechanics research but the literature also documents that still no reliable information on $L/D=f(\mathrm{Re})$ is available.

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Table 1 Summary of publications and results of previous investigations of laminar pipe flow entrance length

No.	Authors	Year	Method	L/D
1	Boussinesq [8]	1891	Analytical, two-zone approach	0.065 Re
2	Schiller [2]	1922	Integral approach, assuming a parabolic velocity distribution within the boundary layer	0.0288 Re
3	Atkinson and Goldstein [10]	1938	Analytical, generalization of the Blasius's solution of the flat plate	0.065 Re
4	Langhaar [3]	1942	Analytical, linearized the momentum equation	0.0575 Re
5	Nikuradse [9]	1950	Experiment	0.0625 Re
6	Siegel [11]	1953	Analytical, integral approach, assuming a cubic velocity distribution within the boundary layer	0.030 Re
7	Bogue [12]	1959	Analytical, integral approach, assuming a quartic velocity distribution within the boundary layer	0.0288 Re
8	Campbell and Slattery [14]	1963	Analytical, applying a plus viscous dissipation of energy term to the entire flow	0.0675 Re
9	Collins and Schowalter [15]	1963	Analytical, developing the work of Boussinesq	0.061 Re
10	Sparrow et al. [4]	1964	Analytical, utilizing a stretched axial coordinate and introducing a mechanical energy equation	0.056 Re
11	Hornbeck [16]	1964	Numerical, initial value problems	0.057 Re
12	McComas and Eckert [17]	1965	Experimental, by relating the pressure drop to distance from the pipe entry	(0.03–0.035)Re
13	Christiansen and Lemmon [18]	1965	Numerical, by obtaining a comparative numerical solution of the momentum equation	0.0555 Re
14	Vrentas et al. [19]	1966	Numerical, boundary value problem	0.056 Re
15	McComas [20]	1967	Analytical, general method depends only on knowledge of the fully developed velocity profile	0.026 Re
16	Friedmann et al. [21]	1968	Numerical, assuming that the initial velocity profiles are concave in the core rather than uniform	0.056 Re
17	Atkinson et al. [22]	1969	Numerical, finite element method	0.59+0.056 Re
18	Lew and Fung [6]	1970	Analytical, two sets of eigenfunctions	0.065 Re(Re$<$1) 0.08 Re(Re$>$50)
19	Fargie and Martin [23]	1971	Numerical, simplified the momentum equations by employing differential and integral form	(0.049–0.068)Re
20	Chen [24]	1973	Analytical, integral momentum method	(0.052–0.068)Re
21	Gupta [25]	1977	Numerical	0.0675 Re
22	Mohanty and Asthana [7]	1979	Analytical, two-zone approach to the inlet and filled regions	0.075 Re

The number of various references cited in the above-mentioned review publications readily suggests that research on the development length for laminar pipe and channel flows is well known and the results are well documented, so that the literature survey in the present paper can concentrate only on tabulating the final results. A summary of investigations, provided in Table 1, clearly indicates that most of the studies on the development length for laminar pipe flows were carried out analytically, with some approximations. The investigations were based either on the integral form of the boundary layer equations or on the zone method, treating the flow close to the entrance of the pipe and in the far-field region separately. Results were also obtained however, by experimental and numerical investigations, as can be seen in the methods column of Table 1.

The results presented in Table 1 are not only widely scattered but also show that different relationships are suggested for the normalized development length of laminar pipe flow; i.e., both the relations $L/D=C\,\mathrm{Re}$ and $L/D=C_0+C_1\,\mathrm{Re}$ (only one publication, by Atkinson et al. [22]), correlate the development length as a function of Reynolds number. Although some of the numerical investigations suggest the first of the above relationships to be valid and provide $C\approx 0.0557$ as a consistent value, the deduced relationship $L/D=C\,\mathrm{Re}$ essentially leaves out the fact that diffusion also plays an important role in the axial direction, particularly at lower Reynolds number, and this yields: $L/D=C_0+C_1\,\mathrm{Re}$ as the "correct relationship" for L/D. This relationship was reported only by Atkinson et al. [22].

In order to obtain a better overview of the results summarized in Table 1, Fig. 1 provides information on the scatter of the entrance length coefficient C, determined by various investigators. The order of the data in this figure is presented, from left to right, in accordance with the year of publication. The results clearly show that, in general, $0.025\leqslant C\leqslant 0.08$.

Regarding the constant, C_0, in the relationship $L/D=C_0+C_1\,\mathrm{Re}$, it is worth noting that in the literature only Atkinson et al. [22] claimed that this constant is essential in the relationship for L/D. This might seem surprising since simple considerations, based on scale analysis, such as that already presented in Sec. 1, suggest that the following relations must hold:

$$\lim_{\mathrm{Re}\to 0}\left(\frac{L}{D}\right)=C_0 \text{ and } \lim_{\mathrm{Re}\to\infty}\left(\frac{L}{D}\right)=C_1\,\mathrm{Re} \tag{1}$$

All these findings from the literature regarding the development lengths of laminar pipe and channel flows are unsatisfactory and

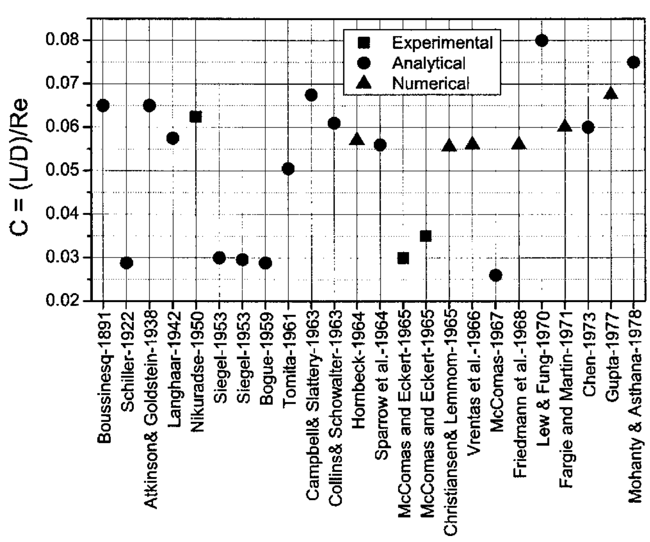

Fig. 1 The constants *C* relationship between the Reynolds number and the ratio of development length and pipe diameter

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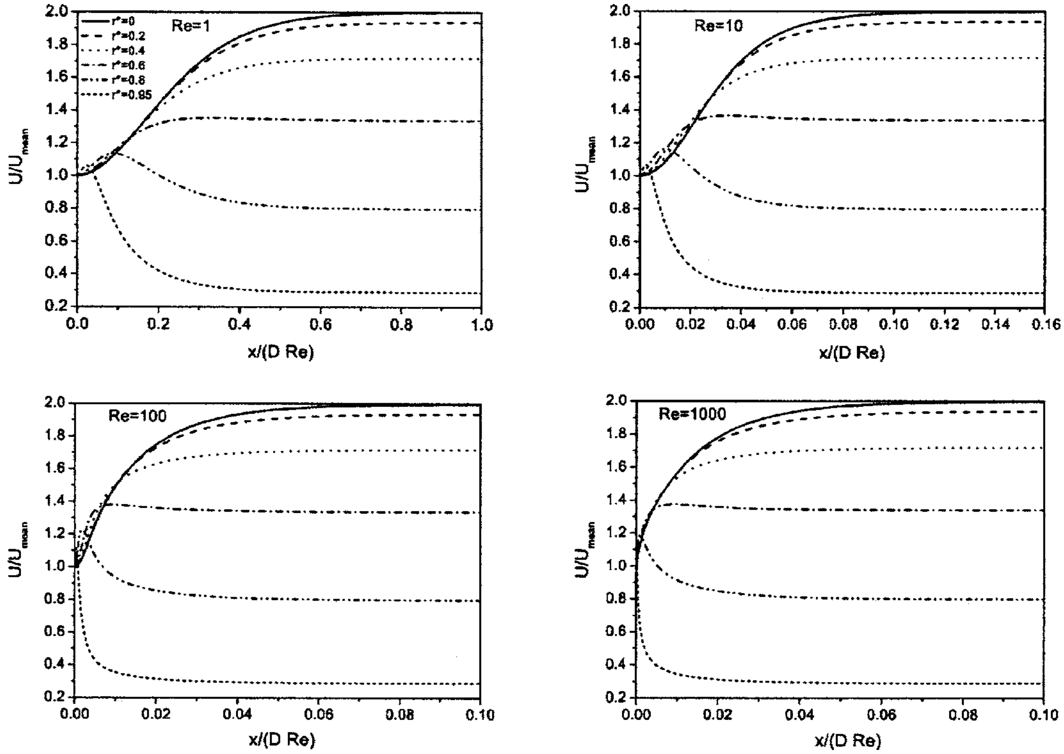

Fig. 2 Axial development of velocities at different radial positions for Re=1, 10, 100, and 1000

suggest that combined numerical, experimental and analytical (if possible) studies are essential to clarify why differences are obtained in the values of C_0 and C_1 in the L/D relationship $L/D = C_0 + C_1$ Re. Such studies and corresponding results are provided in the present paper [13,26].

3 Numerical Investigations

3.1 Equations, Boundary Condition, and Solution Procedure. The present investigations of the development of laminar pipe flow were carried out to yield numerically based data on the development length. Since, for the present investigations, a steady, laminar and developing flow was considered without swirl ($U_\theta=0$) and fluid properties were assumed to be constant, the governing mass and momentum conservation equations were considered in axi-symmetric coordinates. These equations are as follows:

Continuity equation:

$$\frac{1}{r}\frac{\partial}{\partial r}(\rho r U_r) + \frac{\partial}{\partial z}(\rho U_z) = 0 \tag{2}$$

Axial momentum equation:

$$\rho\left(U_r\frac{\partial U_z}{\partial r} + U_z\frac{\partial U_z}{\partial z}\right) = -\frac{\partial P}{\partial z} + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial U_z}{\partial r}\right) + \frac{\partial^2 U_z}{\partial z^2}\right] \tag{3}$$

Radial momentum equation:

$$\rho\left(U_r\frac{\partial U_r}{\partial r} + U_z\frac{\partial U_r}{\partial z}\right) = -\frac{\partial P}{\partial r} + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(rU_r)\right) + \frac{\partial^2 U_r}{\partial z^2}\right] \tag{4}$$

To look for a generally applicable numerical solution, all coordinates in the above equations were made dimensionless with respect to the diameter of the pipe, whereas the velocities were normalized to the constant inlet velocity (U_0), which was considered to be uniform throughout the inlet cross section. The pressure was normalized with respect to ρU_0^2.

The dimensionless governing equations were discretized by a finite (control) volume approach similar to that suggested by Ferziger and Perić [27], i.e., with a method that employs collocated variables, using the SIMPLE algorithm imposed by Patankar [28] for the pressure solution. The algebraic equations resulting from the discretization process were solved by the "Strongly Implicit Procedure" (SIP) proposed by Stone [29].

The boundary conditions before normalization, applied to the present problem, may be summarized as follows:

(1) At $z=0$, i.e., at the inlet, $U_z=U_0$ and $U_r=0$ for every r.
(2) At $z=L_z$, the axial diffusion in both of the momentum equations was neglected, i.e., $\partial^2 U_r/\partial z^2 = \partial^2 U_z/\partial z^2 = 0$ for every r.
(3) At $r=0$, i.e., on the centerline of the pipe, a symmetry condition was imposed, i.e., $U_r = \partial U_z/\partial r = 0$ for every z.
(4) At $r=D/2$, i.e., on the walls of the pipe, no slip condition was assumed, i.e. $U_r=U_z=0$ for every z.

With the above boundary conditions, the discretized set of differential equations was solved in a manner now well known in numerical fluid mechanics. The development length was defined as the distance from the inlet where the dimensionless centerline velocity reaches a value of 1.98 which is 1% lower than its fully developed value. In order to ensure that the development of centerline velocity actually implies true establishment of the fully developed profile in the entire cross section of the pipe, the variation of velocities at different radial positions as a function of axial distance are presented in Fig. 2 for different Reynolds numbers. The figure clearly shows that the velocities at the other radial locations attains their fully developed value much earlier than the centerline.

For the two-dimensional channel flows, the set of equations that has been solved is as follows:

Continuity equation:

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Table 2 Grid independence study for circular pipe flow [$C_1 = L/(D\,\mathrm{Re})$ for different grid spacings]

Re	100×20	100×40	200×40	200×80	400×80
10	0.09256	0.08881	0.08880	0.08784	0.08783
100	0.06279	0.05920	0.05922	0.05844	0.05844
1000	0.06077	0.05728	0.05723	0.05648	0.05648
4000	0.06028	0.05674	0.05677	0.05610	0.05610

$$\frac{\partial}{\partial x}(\rho U_x) + \frac{\partial}{\partial y}(\rho U_y) = 0 \tag{5}$$

Axial flow momentum equation:

$$\rho\left(U_x\frac{\partial U_x}{\partial x} + U_y\frac{\partial U_x}{\partial y}\right) = -\frac{\partial P}{\partial x} + \mu\left[\frac{\partial}{\partial x}\left(\frac{\partial U_x}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial U_x}{\partial y}\right)\right] \tag{6}$$

Transverse flow momentum equation:

$$\rho\left(U_x\frac{\partial U_y}{\partial x} + U_y\frac{\partial U_y}{\partial y}\right) = -\frac{\partial P}{\partial y} + \mu\left[\frac{\partial}{\partial x}\left(\frac{\partial U_y}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial U_y}{\partial y}\right)\right] \tag{7}$$

The boundary conditions employed for the two-dimensional channel flows are similar to those imposed for flow through pipes:

(1) At $x=0$, i.e., at the inlet, $U_x = U_0$ and $U_y = 0$ for every y.
(2) At $x = L_x$, the axial diffusion in both of the momentum equations was neglected, i.e., $\partial^2 U_x/\partial x^2 = \partial^2 U_y/\partial x^2 = 0$ for every y.
(3) At $y=0$, i.e., on the centerline of the channel, a symmetry condition was imposed, i.e. $\partial U_x/\partial y = U_y = 0$ for every x.
(4) At $y=b$, i.e., on the walls of the channel, no slip condition was assumed, i.e., $U_x = U_y = 0$ for every x.

Again, a finite volume method was employed to solve this set of equations and development length was computed in a similar manner, as described for pipe flows.

For all the computations, presented in the paper, uniform grid was chosen in the cross-stream direction (r or y, as the case may be) and the numerical grid in the axial direction was varied in GP series, with a common ratio of 1.05. In order to check the accuracy of the present results, numerical computations were carried out for different grid spacing, starting from 100 (in the axial direction) ×20 (in the radial or cross-stream direction) to 400×80 for various Reynolds number. The development lengths, in the form of $C_1 = L/(D\,\mathrm{Re})$, were monitored for each of the grids and for Reynolds number varying from 10 to 4000. The results for circular pipe are shown in Table 2. It is obvious from the table that virtually no variation was observed in the results for 200×80 and 400×80 grid up to 5 significant digits for entire range of Reynolds number. Since the emphasis in the present paper is to obtain very accurate computational data, 200×80 grid was chosen for all the final computations. Similar study for channel flow, which is not presented in the paper, yielded the same grid spacing.

3.2 Predictions and Results. As stated in the Introduction, it was the aim of the present investigations to clarify some of the widespread and partially contradictory information that is available in the literature on the development lengths of laminar pipe and channel flows. If only the convective velocity is taken into account in the flow direction and the diffusion velocity is considered only across the flow for scaling, the following order of magnitude considerations result:

$$U_z = C^* U_0 \text{ and } t_{\mathrm{diff}} = C^+\frac{D^2}{\nu} \Rightarrow \frac{L}{D} = \underbrace{C^* C^+}_{C}\frac{U_0 D}{\nu} \tag{8}$$

from which one readily obtains the high Reynolds number limit of (L/D) behavior:

$$\frac{L}{D} = C\,\mathrm{Re} \tag{9}$$

Postulating this relationship to be valid also for smaller Reynolds numbers, a dependence of C on Re must result. The variation of C [$=L/(D\,\mathrm{Re})$] with Re was deduced from numerical studies on the development length of laminar pipe flow and is shown in Fig. 3. These results account for some of the scatter of C values reported in Sec. 2, i.e., wrong correlation [Eq. (9)] is fitted to the data.

The low Reynolds number limit of $L/D = f(\mathrm{Re})$ can be derived by considering only the diffusion velocity both in the axial direction and in the cross-flow direction. Such consideration yields the following relationships;

$$U_z = C^*\frac{\nu}{D} \text{ and } t_{\mathrm{diff}} = C^+\frac{D^2}{\nu} \Rightarrow \frac{L}{D} = C^* C^+ = C_0 \tag{10}$$

If this consideration for the low Reynolds number limit is not taken into account, it is obvious that the constant C in Eq. (9) must increase with decreasing Reynolds number (see Fig. 3).

In the entire Reynolds number range, $0 < \mathrm{Re} < \infty$, the above order of magnitude consideration can be extended by taking both the convection and the diffusion velocities in the flow direction into account, yielding

$$U_z = \left(C_0^*\frac{\nu}{D} + C_1^* U_0\right) \text{ and } t_{\mathrm{diff}} = C^+\frac{D^2}{\nu} \tag{11}$$

from which the development length may be estimated as

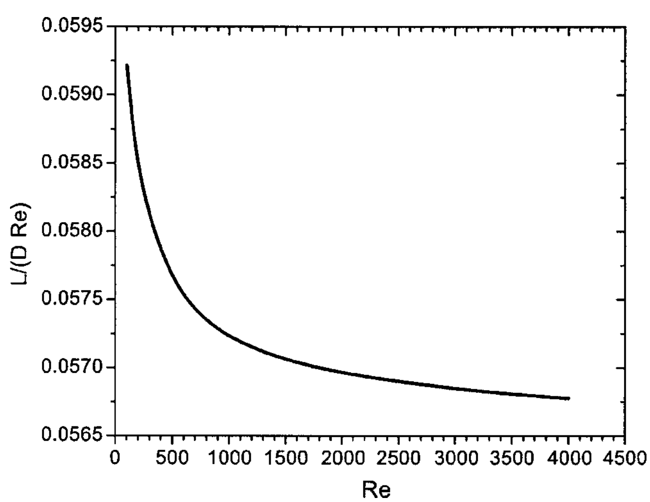

Fig. 3 Variation of constant *C* with Reynolds number of the pipe flow

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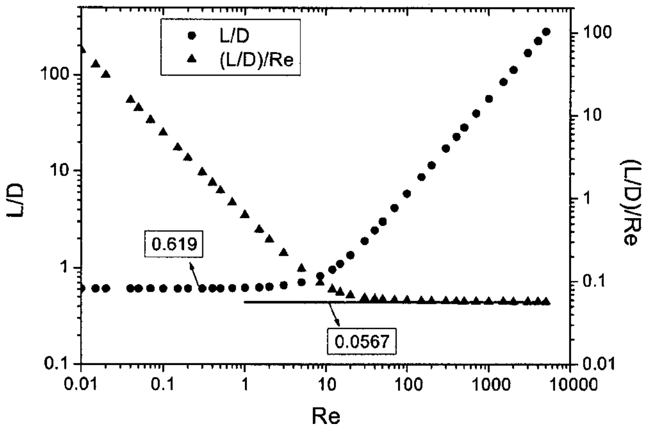

Fig. 4 Numerical results for L/D plotted to deduce C_0 and C_1 in the L/D relationship

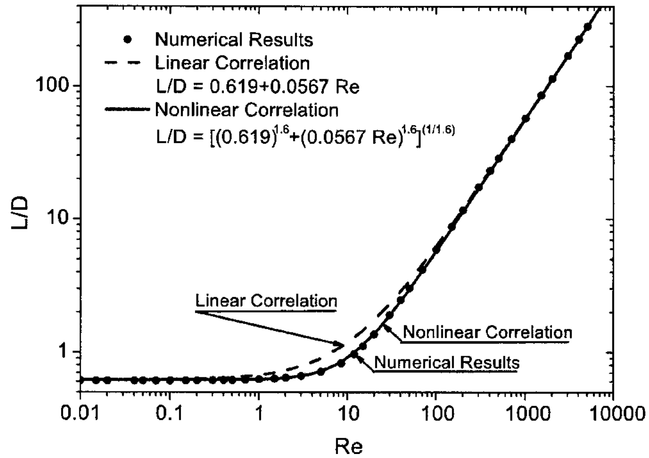

Fig. 5 Predicted development length results and comparison with the convection diffusion relationship in Eq. (14) for laminar pipe flow

$$L = U_z t_{\rm diff} = D(C_0^* C^+ + C_1^* C^+ \,{\rm Re}) \Rightarrow \frac{L}{D} = C_0 + C_1\,{\rm Re} \tag{12}$$

Fitting this function to the numerical results for L/D, one readily obtains

$$\lim_{{\rm Re}\to 0}\left(\frac{L}{D}\right) = C_0 = 0.619 \tag{13}$$

The high Re numerical data are fitted to $L/(D\,{\rm Re})$ to yield

$$\lim_{{\rm Re}\to \infty}\left(\frac{L}{D\,{\rm Re}}\right) = C_1 = 0.0567 \tag{14}$$

The numerical data for L/D and $L/(D\,{\rm Re})$ are presented in Fig. 4 and also clearly indicate the limiting values given in Eq. (13) for low Reynolds numbers and in Eq. (14) for high Reynolds numbers. It may be recognized that the low Reynolds number limit takes only the diffusive momentum transport into account, whereas the high Reynolds number limit considers only the convective transport in the flow direction and the diffusive transport in the cross-flow direction.

Taking the computational results and their limiting behavior into account raises a question regarding the validity of the deduced relationship

$$\frac{L}{D} = 0.619 + 0.0567\,{\rm Re} \tag{15}$$

in the range $1 \leqslant {\rm Re} \leqslant 100$. In this range of Reynolds number, the nonlinearity of the convection terms in the momentum transport equation yields conditions that do not permit L/D to be simply computed by the addition of C_0 and C_1 Re terms. This can readily be seen from Fig. 5, which shows the predicted L/D correlation given by Eq. (15) (dashed line). The actual computations are given by circles. A comparison of both results suggests that, for a given Reynolds number, the actual development length of a laminar pipe flow is shorter than the length calculated from the correlation in Eq. (15). At this point, since the asymptotic behaviors of L/D in both low and high Reynolds number regimes is known, the suggestions of Churchill and Usagi [30] may be employed to correlate the data as follows:

$$\frac{L}{D} = [(0.619)^{1.6} + (0.0567\,{\rm Re})^{1.6}]^{1/1.6} \tag{16}$$

The above correlation is also shown in Fig. 5, which predicts the numerical data within a maximum error of 3%.

The above-described numerical investigations of the development length for laminar pipe flow were also repeated for laminar, plane channel flow solving Eqs. (5)–(7) and the corresponding boundary conditions. As expected, the numerical computation yielded similar results. The high and low Reynolds number limits turned out to be as follows:

$$\lim_{{\rm Re}\to 0}\left(\frac{L}{D}\right) = C_0 = 0.631 \tag{17}$$

$$\lim_{{\rm Re}\to \infty}\left(\frac{L}{D\,{\rm Re}}\right) = C_1 = 0.0442 \tag{18}$$

Atkinson et al. [10] also computed these constants and found them to be C_0=0.625 and C_1=0.044, which are in close agreement to the findings given above. However, as Fig. 6 shows, the corresponding correlation

$$L/D = 0.631 + 0.0442\,{\rm Re} \tag{19}$$

again yields, for a given Reynolds number range, a development length for laminar plane channel flow that is longer than the numerically computed length. The reasons for this are principally the same as those given for the difference for the laminar pipe flow. Here also, one can fit the numerical data with the following correlation:

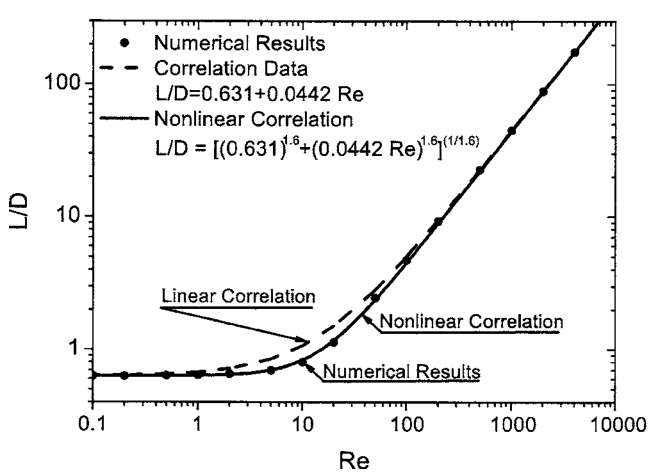

Fig. 6 Predicted development length results and comparison with convection diffusion relationship for laminar channel flow

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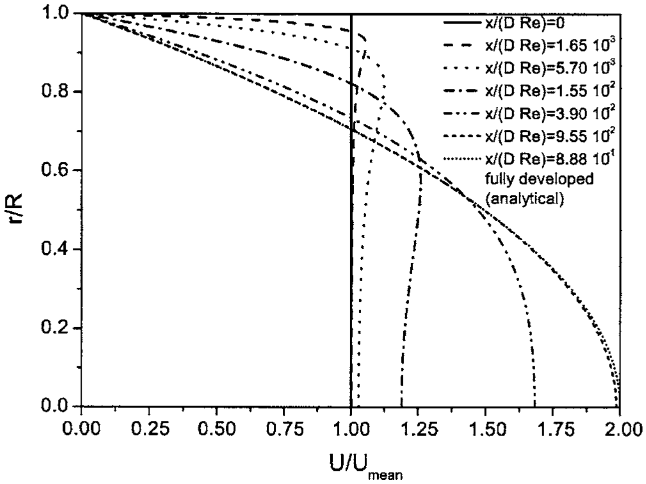

Fig. 7 Axial development of velocity profile for Re=10

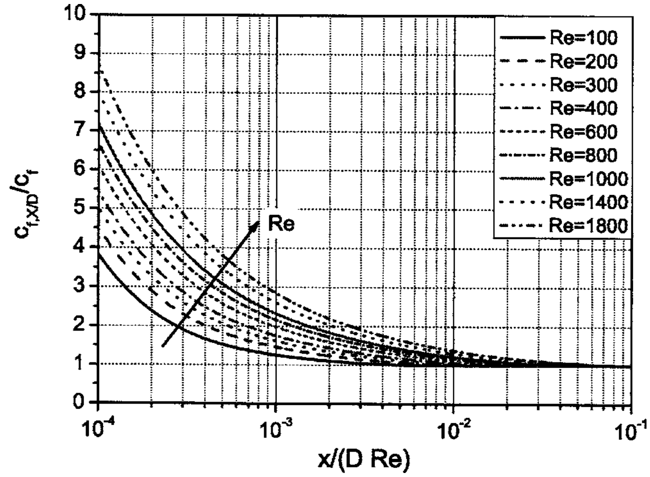

Fig. 8 Change of local skin friction factor through the entrance region of the pipe

$$\frac{L}{D} = [(0.631)^{1.6} + (0.0442\,\mathrm{Re})^{1.6}]^{1/1.6} \quad (20)$$

The above correlation is also shown in Fig. 6, which fits the computational data within 3.8%. It is also interesting that in the correlations for both pipe and channel flows [see Eqs. (16) and (20)], although the constants C_0 and C_1 vary, the exponent is the same and is equal to 1.6.

Furthermore, the change of skin friction factor $(c_{f,x/D})$ in the entrance region investigated for pipe flow. For this calculation much finer grid was used especially through the entrance region for better resolution. The results are shown for varying Re numbers in Fig. 7. The local skin friction factor $(c_{f,x/D})$ was normalized with c_f $(=16/\mathrm{Re})$ and location (x/D) normalized with Re. Very sharp decrease in $c_{f,x/D}$ through the entrance region can be seen in Fig. 7.

4 Experimental and Analytical Investigations

As Table 1 shows, most of the investigations in the past, carried out to yield the development length for laminar pipe and channel flows, were carried out experimentally or analytically. The results also show (see Fig. 1) that the research, over the years, did not yield a convergence to a certain relationship for L/D for either pipe flows or channel flows. This reflects the fact that there are insurmountable difficulties in determining the development length L/D of laminar pipe and channel flows experimentally and/or analytically. To assess these difficulties, we carried out experimental investigations using an air test rig and a well set-up pipe flow test section. In the air test rig, hot-wire measurements were performed to yield the axial velocity at the exit of a pipe of given L/D (essentially pipes of fixed lengths-to-diameter ratio were employed). From a preset mass flow rate, $\dot{m}$=const and the volume flow rate was computed as $\dot{V}=\dot{m}/\rho$, where ρ was calculated from the measured air pressure and the measured air temperature of the flowing fluid using the ideal gas law. The known volume flow rate permitted the cross-sectional mean flow velocity to be determined and from this the expected maximum velocity, assuming a parabolic velocity profile, on the pipe axis was obtained:

$$U_{\mathrm{av}} = \frac{4\dot{V}}{\pi R^2} \Rightarrow U_{\max} = 2U_{\mathrm{av}} \quad (21)$$

Changing experimentally the set mass flow rate $\dot{m}$ in steps permitted the velocity on the axis to be determined at the pregiven distance L/D by means of a well-calibrated hot-wire system. When this velocity agreed within 2% with the corresponding calculated maximum velocity, the flow was assumed to be fully developed. Carrying out velocity measurements for L/D=74.67, 141.33, 208, and 274.67 permitted L/D as a function (Re) to be experimentally determined.

The L/D values obtained in this manner might be called an asymptotic assessment of L/D. These experiments revealed that any small measurement inaccuracies eventually yielded large variations of the measured Reynolds number for which a measured $U_{\max}$ was obtainable to be close to 2% of the $U_{\max}$ calculated from the preset $\dot{m}$. This fact and the strong dependence of C_1 in the expression of L/D on asymptotically measured Reynolds number might have caused, in the past, the large scatter in the experimentally determined L/D values indicated in both Table 1 and in Fig. 1.

Analytical studies were also carried out of the development length of laminar pipe and channel flows. Strictly, owing to the presence of non-linear terms in the convective transport of the momentum equations, no analytical solution is possible without further assumptions being introduced. Hence all the solution attempts available in the literature may be considered to be only "semianalytical." In order to obtain such semianalytical solutions for both the development length of the laminar pipe and channel flows, two different approaches were adopted. In the first approach, which was used by many of the early researchers, the integral momentum equation with boundary layer approximation was adopted. In this approach, the free stream velocity was considered to be the centerline velocity, which varies with the axial distance, obeying the appropriate integral mass conservation equation. With in the assumed boundary layer developing along the pipe, both parabolic and cubic velocity distributions were considered, i.e., were assumed to be valid for the L/D calculation. However, such solutions are valid only for the high Reynolds number regime, since they all neglect the contribution of the axial diffusion, which is otherwise important in the lower Reynolds number regime. Hence the applied integral momentum equation solutions can only provide an acceptable dependence of L/D on Re as given in Eq. (9). The velocity profiles at various axial locations for Re =10, are presented in Fig. 8. It is observed from the figure that at locations near to the entrance, velocity overshoots occur close to the wall. These overshoots were not taken into account in the semianalytical treatment employing boundary layer approximations, since such inflection points in the velocity distribution could not be captured by the employed parabolic or cubic profiles. Hence the evaluation of the constants, appearing in Eq. (11), by this method is also liable to be incorrect.

In the second approach, a starting flow solution was tried, where the initial condition is given by a uniform velocity distribution throughout the entire pipe cross section. In this problem the

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governing equations were considered to be the same as those applicable for transient flow in the fully developed section. It was assumed that the transient term in this situation plays a similar role to that of the convection terms for steady developing flows with uniform inlet velocity. From the transient velocity distribution (which may be obtained in terms of an infinite series of cosine and sine terms), the time required to achieve the steady-state condition (to be precise, when the velocity distribution reaches 99% of that of the steady fully developed solution) was calculated. Using this time and the average axial velocity, the distance travelled by an average fluid particle to reach a steady state was obtained. This distance was considered to be development length, and this can only be correct if the assumption regarding the convection term is correct. However, it was observed that the solution not only provides the dependence given in Eq. (9), but also overpredicts the development length by a factor of 2. Hence, from the present effort to obtain an analytical solution for the development length through pipes, it may be concluded that no analytical (or even semianalytical) solution is possible which correctly predicts the dependence of L/D in both the lower and higher Reynolds number regimes.

5 Conclusions and Final Remarks

Investigations of the development length of laminar pipe and channel flows have been carried out extensively and various reports of relevant work are available in the literature. In spite of this, no analytical expressions exist that correctly describe the development length for both types of flow. This encouraged the present work employing numerical, experimental and analytical methods. Only the numerical investigations were successful, yielding the following analytical relationships for the correlation length:

$$\text{Pipe flow: } \frac{L}{D} = [(0.619)^{1.6} + (0.0567\,\mathrm{Re})^{1.6}]^{1/1.6} \tag{22}$$

$$\text{Channel flow: } \frac{L}{D} = [(0.631)^{1.6} + (0.0442\,\mathrm{Re})^{1.6}]^{1/1.6} \tag{23}$$

Mostly in the literature the correlation $L/D = C\,\mathrm{Re}$ was used. This correlation neglects the diffusion transport in axial direction and in the paper it is shown that it should be replaced by $L/D = C_0 + C_1\,\mathrm{Re}$. However, this correlation does not correctly describe the development length in the region $1 \leqslant \mathrm{Re} \leqslant 100$. Hence the above analytical correlations are suggested to compute L/D for channel and pipe flows of different Reynolds numbers.

The authors feel that the results presented in this paper close the long ongoing research work on the development length of laminar pipe and plane channel flows.

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